BTS–I–11.19–1454 Reg. No.

C

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY SCHOOL OF ENGINEERING, KOCHI 682 022 LIBRARY

B.Tech. Degree I Semester Examination November 2019

CE/EE/ME/SE 19-200-0105 A BASIC ELECTRONICS ENGINEERING

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(8 \times 3 = 24)$

I. (a) Explain how conduction takes place in semiconductor materials.

(b) Explain depletion width and barrier potential of pn junction.

(c) Write notes on the effect of variation of temperature on power rating of a diode.

(d) Explain zener break down and avalanche breakdown.

(e) In a transistor with base current 10 μA and β=100, find out collector current, emitter current and α.

(f) Explain the working of an npn transistor.

(g) Briefly explain diffusion and epitaxial growth with diagrams.

(h) Define the types of ICs based on its manufacturing techniques.

PART B

$(4 \times 12 = 48)$

II. (a) Explain the electrical behavior of conductors, semiconductors and insulators with the help of their energy band diagrams. (6)

(b) Discuss the effect of forward and reverse biasing of pn junction on the width of the depletion region. (4)

(c) Determine the levels of reverse saturation current at temperatures of 35^0C and 45^0C for a PN-junction which has reverse saturation current of 30nA at 25^0C. (2)

OR

III. (a) Explain the effect of temperature on the resistance of conductors and semiconductors. (4)

(b) Define doping and explain how p type semiconductors are formed. (5)

(c) A silicon pn junction has a reverse saturation current of 30 nA at a temperature of 300^oK. Calculate the junction current when the applied voltage is (i) 0.7V forward bias (ii) 10V reverse bias. (3)

IV. (a) Illustrate the forward and reverse characteristics of a pn junction diode and define its dynamic resistance. (8)

(b) A silicon diode with a 0.7V forward voltage drop at 25^0C is to be operated with a constant forward current up to a temperature of 100^0C. Calculate the diode forward voltage drop at 100^0C. Also determine the junction dynamic resistance at 25^0C and at 100^0C, if the forward current is 26 mA. (4)

OR

V. (a) Explain the working of half wave rectifier and bridge rectifier. (8)

(b) Write the DC load line equation and plot the DC load line of a forward biased diode connected in series with a DC voltage E and a resistance R. Explain how the operating point of the circuit is located graphically. (4)

(P.T.O.)

VI.	(a)	Explain the working of transistor as a voltage amplifier.	(6)
	(b)	Explain the input and output characteristics of CE transistor configuration.	(6)

OR

VII.	(a)	Illustrate the working of CRO with detailed diagrams.	(8)
	(b)	Explain the working of transistor as a switch.	(4)
VIII.	(a)	Briefly discuss the constructional requirements of transistors with (i) Good current gain (ii) High frequency response (iii) High breakdown voltage	(6)
	(b)	Explain construction of alloy , micro alloy and micro alloy diffused transistors.	(6)

OR

IX.	(a)	Discuss the fabrication of integrated circuit transistors, resistors and capacitors.	(6)
	(b)	Describe the construction of diffused mesa and annular transistor.	(6)

***